

Speaker Notes

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Slides: 34 · **Duration:** 28 min · **Venue:** Faculty of Informatics, Masaryk University (FI MUNI)

11. Practicum in Operating Systems

On screen:

PB158 · Lecture 1

Course orientation

Ondřej (Ondra) Krajíček me@ondrejkrasicek.com · [linkedin.com/in/OndrejKrajicek](https://www.linkedin.com/in/OndrejKrajicek) Built at render time

Good morning. This is PB158, Practicum in Operating Systems, and this is the first of two lectures today. This one runs about half an hour. It carries no operating-system content at all. It tells you who teaches this course, what it covers and how it grades you. It also tells you what must work on your laptop before the first lab. Then we take a short break, and the second lecture starts the actual subject. I want to be honest about the shape of the next thirty minutes. This is the administrative half, and I am going to argue that the administration is not separate from the subject. How a course measures you tells you what it thinks the subject is.

2 2. What we'll cover

On screen:

Lecture 1

This course grades you on what you can see a system do. It does not grade you on what you can recite about it. [speculative]

1	Who and why	<i>The gap a practicum fills beside your theory courses</i>
2	The topics	<i>Eight topic areas, and what each one lets you do</i>
3	Assessment	<i>Two components, three assignments, real dates, who grades them</i>
4	Reading	<i>Four people who write well about this material</i>
5	Your machine	<i>What to install before the first lab</i>

The payoff is part 3. Passing an assignment removes its subject from your exam. [speculative]

That first line is the thesis of the whole term. I argue for it in part one. Five parts. The order matters. Part three is the one you came for. I will tell you the headline now rather than make you wait for it. Three homework assignments this semester. They are pass or fail, and there are no points. If your team passes an assignment, the subject of that assignment is removed from your exam. Pass all three and only an irreducible core is left. I will give you the exact dates and the exact mechanism when we get there. Part one is four minutes on why this course exists, because everything in part three follows from it. If the assessment looks arbitrary, it is because you have not heard part one yet. [speculative]

3 3. Why this course exists

On screen:

Part I

The gap between what you can define and what you have seen.

Part one. Four minutes. Two things to establish here: who is responsible for this course, and what the course is for. The second one is the thesis of the whole term. I want to state it clearly. You should not have to infer it from the rules in part three.

4 4. Who runs this course

On screen:

Part I · Staff

Ondřej Krajiček	<code>krajicek@mail.muni.cz</code> — lecturer
Co-lecturer	named in IS MUNI, with contact details, once the course opens

Assisting with the course: **three teaching assistants**, whom you will meet in the labs. IS MUNI carries the roster.

Material comes **from IS MUNI**. Work goes **to FI MUNI GitLab**, at `gitlab.fi.muni.cz`. Two systems, two directions. [speculative]

Write to either of us. You do not need a reason, and you do not need to wait for a seminar.

A co-lecturer and I teach this course together. Three teaching assistants join us in the labs; IS MUNI carries the roster, so I am not reading names at you. Two logistics first. Material comes from IS MUNI. Slides, the handout and the assignment briefs all live there. Work goes the other way, to the faculty GitLab. Two systems, two directions. Then two points about the course itself. It is mandatory, so some of you did not choose it. I would rather earn your attention than assume it. And write to us early. There are fewer than ten of you, so interrupt me. [speculative]

5 5. You can define a system call. You have not seen one.

On screen:

Part I · The gap

What you have You can write Python. You can write C. You can define a process, a system call and virtual memory. [speculative]

What you have not You have never attached a tracer to a running program and read what it actually did. [speculative]

This is not a gap in you. It is a gap in the curriculum, and it dates from a licence change in 1979. [S5]

[S5] Lessons Learned from 30 Years of MINIX — Tanenbaum, CACM (2016). <https://www.cs.vu.nl/~ast/Publications/Papers/cacm-2016.pdf>

Let me describe the room as I understand it. You write Python and you write C. You have had operating systems as theory. You can define a system call, a process, virtual memory, a page. None of you has ever attached a debugger or a tracer to a running program. That is a strange combination if you think about it. You can define the thing and you have never watched it happen. I want to be clear that this is not a criticism, and it is not an accident either. It is the normal output of a computer science curriculum. [speculative] The reason is historical. It goes back to a licence change in nineteen seventy nine. [S5]

6 6. Operating systems went back to theory in 1979

On screen:

Part I · History

In 1977 John Lions wrote a line-by-line commentary on the UNIX version 6 kernel. Hundreds of universities taught from it. [S5]

The 1979 AT&T release changed the licence and forbade teaching the source. Courses went back to theory-only mode or moved to toy simulators. [S5]

The teaching stayed theoretical [S5]. A disabled DTrace probe costs nothing, which is "what made DTrace safe to ship enabled on production systems" [V23]. You will attach one to a running program. [speculative]

[S5] Lessons Learned from 30 Years of MINIX — Tanenbaum, CACM (2016). <https://www.cs.vu.nl/~ast/Publications/Papers/cacm-2016.pdf> · [V23] Observing a Running System

Andrew Tanenbaum tells this story in his retrospective on MINIX. John Lions wrote a commentary on the sixth edition UNIX kernel in nineteen seventy seven, and it became the textbook of hundreds of universities. Then the nineteen seventy nine release shipped with a licence that forbade writing about the source or teaching it. Tanenbaum writes that operating systems courses went back to theory-only mode, or had to use toy simulators. He wrote MINIX to fix exactly that. Now, the interesting part is what

happened next. The teaching stayed theoretical. DTrace patches the running system only when you enable a probe. A disabled probe costs nothing. That is what made DTrace safe to ship enabled on production machines. DTrace is one example, and topic three gives the cost of the others. This course puts you in front of a tracer of that kind, attached to a running program. You have the theory already. What you do not have is the habit of looking. [S5] [V23] [speculative]

7 7. Looking and reciting are different skills

On screen:

Part I · Thesis

Reciting

You answer a question somebody already wrote down. A monitoring dashboard does this: "it answers questions you already knew to ask" [V2]

Looking

You ask a question nobody wrote down in advance, and the running system answers it. The field calls this observability: "the ability to investigate a system by asking any question about its behavior without needing to predict those questions in advance" [V1]

This course grades the second one. You can already recite more than you can predict. [speculative]

Here is the thesis of the course. I want to give you the technical version rather than a slogan. The two words for this already exist. The first is monitoring. Monitoring answers questions somebody already knew to ask. Somebody chose the metric, set the threshold, wrote the alert. A written theory test is monitoring. The second word is observability. The standard definition is the ability to investigate a system by asking any question about its behaviour, without needing to predict those questions in advance. That is what this course grades. And I claim you can recite more than you can predict. Here is a thirty second exercise to test that claim. Nobody has to answer aloud. [V1] [V2]

8 8. Thirty seconds, silently, to yourself

On screen:

Part I · Test it

In your head, say the steps between a Python `f.write()` returning and the bytes being on the disk.

Rozenblit and Keil measured what happens next. People rate their own understanding lower after they try to explain the thing than before. [S1]

If your explanation had a gap, that gap is the course. Nobody is graded on this, and nobody has to say it aloud. [speculative]

[S1] The misunderstood limits of folk science: an illusion of explanatory depth — Rozenblit & Keil, Cognitive Science 26 (2002). cogdevlab.yale.edu

Take thirty seconds. Do not say anything aloud. In your own head, state the steps between a Python write call returning and your bytes reaching the disk. Name each layer. I will wait. Right. Some of you got two steps and stopped. That result has a name and a measurement. Rozenblit and Keil called it the illusion of explanatory depth. They asked people to rate their understanding, then to explain the thing, then to rate themselves again. The second rating was consistently lower. The effect is strongest for explanations and weakest for plain facts. That is the shape of a theory-only education. Trying to explain reveals the gap. Hearing about it does not. [S1]

9 9. What a practicum asks of you

On screen:

Part I · Practicum

A lecture course asks you to reproduce an explanation. A practicum asks you to produce an observation and defend it. [speculative]

Crouch and Mazur measured the difference in a physics classroom. Students who only watched a demonstration gained little understanding of the concept it showed. Students who predicted the outcome first understood that concept measurably better. [S2]

So I will not prove the thesis in this lecture. The next lecture opens with a three-line program whose output disappears.

[S2] Classroom demonstrations: learning tools or entertainment? — Crouch, Fagen, Callan & Mazur, Am. J. Phys. 72 (2004). <https://works.swarthmore.edu/fac-physics/203/>

A practicum is not a lecture course with exercises attached. The difference is what it asks you to produce. A lecture course asks you to reproduce somebody else's explanation. A practicum asks you to produce an observation and then defend it. Crouch, Fagen, Callan and Mazur measured this in a physics classroom. Passive watching produced very little understanding of the concept demonstrated. Predicting the outcome first produced a measurable gain in that understanding. Predict, then look. That is the method. Now notice what I have just done. I asserted the thesis of this course and proved nothing. That is deliberate. The next lecture opens with three lines of Python whose output vanishes. Hold me to it. [S2]

10 10. Takeaways

On screen:

Part I · Hold these

1

The course grades looking

Not reciting. The assessment in part 3 is built around that, and so is every topic [speculative]

2

It works downward

From the code you write to the operating system under it. Each topic takes one thing you already use and shows the mechanism [speculative]

Two claims. If you remember nothing else from this half hour, remember these.

Two takeaways from this lecture. First, this course grades you on what you can see a system do. It does not grade you on what you can recite about it. Everything in part three follows from that. Second, the course works downward. It starts from code you already write, in Python and in C, and goes down through the layers to the operating system underneath. Every topic takes one abstraction you use daily, and shows you the mechanism it rests on. Those are the two claims. I will repeat them at the end. Now let me show you the eight topic areas, because the second claim is easier to believe once you see the list. [speculative]

11 11. The course

On screen:

Part II

Eight topic areas, and one idea that carries all of them.

Part two, and it is the longest part of this lecture. Twelve minutes. I am going to name eight topic areas and tell you what each one is about. Before that, one idea, because the eight topics are eight instances of it rather than eight separate subjects.

12 12. One idea carries all eight topics

On screen:

Part II · The idea

Every operating-system interface keeps its promise only under conditions nobody wrote down. [inference]

<i>The contract</i>	<i>The written half says what the call does. The scope clause says when it does that. [V4]</i>
<i>The method</i>	<i>You cannot reason out which layer lies. Record the boundary crossings. Then count them. [V5]</i>

A leak is your program leaving the scope, not the abstraction failing [V4]. A library's documentation is a claim. The system call it issues is a fact [V21]. Trust the trace when they disagree [inference].

[V4] [V5] Where the Abstraction Leaks · [V21] AI Agents as Debugging Operators

Here is the idea. Every abstraction the operating system gives you is a contract with a silent scope clause. The contract is written down. Open a file, write bytes, close it. You have read that part. The scope clause is the part nobody writes down. It says under which conditions the contract holds. Step outside those conditions and you get a leak. The leak is not a bug. It is your program leaving the scope. And here is why this course is built on tools. You cannot deduce which layer is lying. You have to make the boundary crossings visible and count them. One more sentence, and I would put it on the

wall. A library's documentation is a claim. The system call it issues is a fact. [V21] When the two disagree, trust the trace, and read the documentation as the wrong one. [inference] [V4] [V5]

13 13. Topic 1 — I/O: files and the network

On screen:

Part II · Topic 1

<i>What it covers</i>	<i>One file write and one network request, to the device</i>
<i>The general rule</i>	<i>Nearly every fast call returns before the work happens. "Almost every fast OS interface returns when the kernel has taken responsibility for the work, not when the work has happened." [V6]</i>
<i>The divergence</i>	<i>On Linux a socket is a file descriptor. On Windows it is not: "In Winsock applications, a socket descriptor is not a file descriptor and must be used with the Winsock functions." [V7]</i>

The course starts here: following one write down makes a tracer necessary. [speculative]

[V6] Where the Abstraction Leaks · [V7] Sockets and Network I/O

Topic one is input and output, files and the network together. Everyone here has written a file, so this needs no motivating. Here is the rule you meet. Almost every fast operating system interface returns when the kernel has taken responsibility for the work. It does not return when the work has happened. Those are different moments. The gap between them is where the interesting failures live. Files and sockets share a slide because read and write operate on both. The platforms disagree sharply here. On Linux a socket is a file descriptor. On Windows, in Microsoft's own words, it is not. Why start here? Following one write call down is what forces the boundary into view, and what makes a tracer necessary rather than decorative. That is my own reason for the order. [V6] [V7] [speculative]

14 14. Topic 2 — The system-call boundary

On screen:

Part II · Topic 2

What it covers	<i>Where your program stops and the operating system begins, and how you cross that line</i>
The divergence	<i>Linux publishes the system call interface. Windows publishes Win32 and reserves the layer below it [V8]</i>
The correction	<i>"Both are stable at the layer they nominate. The disagreement is about which layer is nominated" [V8]</i>

You will be able to name where the boundary sits on each platform, and say why they chose differently. [speculative]

[V8] Windows and Linux place the stable syscall boundary at different layers — author's working notes, summarised in the handout appendix.

Topic two takes what topic one showed you in one case and makes it general. Where does your program stop and the operating system begin? Linux publishes the system call interface itself as the stable contract. You may issue a system call directly and the kernel promises it will keep working. Windows does not do that. Windows publishes Win32 as the contract and treats the layer below as an implementation detail it reserves the right to change. Now, the easy mistake here is to read that as Linux doing it right. It is not. Both systems are stable at the layer they nominate. The disagreement is about which layer gets nominated, and each choice buys something and costs something. We will work through both. [V8]

15 15. Topic 3 — Observing a running system

On screen:

Part II · Topic 3

What it covers	<code>strace</code> on Linux, NtTrace on Windows, and how to read what they print
The honest limit	A tracer adds code, so it changes the timing it reports [inference]. "A tracer is not a window onto a running program. It is an edit to that program." [V22]
The precedent	A published graduate course "teaches kernel behaviour through DTrace and hardware performance counters" [V9] [S3]

You will be able to state a tracer's cost and its blind spot. [speculative]

[S3] Teaching OS with FreeBSD through Tracing — Neville-Neil et al. (2017). https://www.usenix.org/system/files/login/articles/login_spring17_03_neville-neil.pdf · [V22]

[V25] Tracing cost and blind spot

Topic three is the tools. On Linux that is `strace`. On Windows it is NtTrace, which attaches at a different boundary for the reason we just covered. The part I care most about is the limit. A tracer is not a window onto a running program. It is an edit to that program. [V22] Three consequences follow:

- The tracer changes the timing it reports. [inference]
- The tracer costs real performance. [V22]
- Every tracing mechanism has a blind spot. [V22]

A syscall tracer sees only the work that crosses the boundary. Linux has an interface called `io_uring` that passes requests through shared memory instead, and `strace` never sees those. [V25] You must leave this course able to run your platform's tracer — `strace` on Linux, NtTrace on Windows — and able to state its blind spot. With only the first, I have taught you a command rather than a skill. [speculative] This design is not mine. Neville-Neil and colleagues published a graduate course taught this way. [V9] [S3]

16 16. Topic 4 — Virtual memory and the page cache

On screen:

Part II · Topic 4

What it covers	<i>Where the bytes you wrote went, and why freed memory is still held</i>
The mechanism	<i>An allocation gives you address space, not memory. "A mapping can exist with no physical page behind it. The page is materialized on first touch, by a fault." [V17]</i>
The consequence	<i>One process yields several memory figures, each measuring a different thing. "How much memory does this process use" turns out to be a badly posed question [V17]</i>

You will be able to read a memory figure and say which question it answers. [speculative]

[V17] Virtual Memory and the Page Cache — author's working notes, summarised in the handout appendix.

Topic four. You allocate memory and you think you now have memory. You do not. You have address space. A mapping can exist with no physical page behind it, and the page arrives on first touch, by a fault. A page fault is not an error. It is the normal mechanism by which memory starts existing. Once you know that, several confusing things stop being confusing. Why a program that re-reads the same file is faster on its second run. Why your process monitor disagrees with your program. Why free memory is mostly a bookkeeping category. The skill from this topic is reading a memory number and knowing which question it answers. [V17]

17 17. Topic 5 — Protection, shared memory and IPC

On screen:

Part II · Topic 5

What it covers	<i>How one address space is isolated from another, and every way you deliberately cross that line</i>
The mechanism	<i>"Every load, store, and instruction fetch a process performs is checked before it completes." [V18]</i>
Where it is checked	<i>In hardware, by the memory management unit, against permission bits the kernel wrote [V18]</i>

You will be able to say why a segmentation fault is the protection working rather than failing. [speculative]

[V18] Memory Protection — author's working notes, summarised in the handout appendix.

Topic five. Every load, every store and every instruction fetch your process performs is checked before it completes. Not by the compiler. Not by the language runtime. Not by the kernel code. It is checked in hardware, by the memory management unit, on every access, against permission bits the kernel wrote. That is why isolation is cheap enough to be unconditional. It also reframes something you have all seen. A segmentation fault is not the system breaking. It is the protection working, and telling you so. The second half of the topic covers the deliberate crossing: shared memory, pipes, message queues, signals. [speculative] Isolation is the default, and IPC is how you open one door in it. [V18]

18 18. Topic 6 — Processes, threads and scheduling

On screen:

Part II · Topic 6

What it covers	<i>What the operating system actually schedules, and how language runtimes redefine the thread you think you have</i>
The finding	<i>"A userspace-scheduled unit of concurrency produces no system call when it is created and no system call when it yields" [V19]</i>
Why that matters	<i>"a syscall tracer shows nothing at all where the student expects to see concurrency" [V19]</i>

Here the absence is the finding [inference]. You will be able to read an empty trace as evidence. [speculative]

[V19] Processes, Threads and Scheduling — author's working notes, summarised in the handout appendix.

Topic six contains my favourite result in the course. You write a program using green threads, or goroutines, or async tasks. You attach a tracer, expecting concurrency. You see nothing. No system call when the unit is created, none when it yields. The trace looks like a serial program. Students meet this and assume the tool is broken. It is not. The absence is the finding. It tells you the language runtime has taken scheduling away from the kernel. Your thread is not what the operating system calls a task. Reading an absence as evidence is a genuinely hard skill, and this is where the course teaches it. [V19] [speculative]

19 19. Topic 7 — Post-mortem analysis, with and without an agent

On screen:

Part II · Topic 7

What it covers	<i>Reading what a process left behind, and driving the tools with an AI agent</i>
Why it is separate	<i>Every earlier topic needs a live process. Here it is dead: "A dump is all you get once it is already dead." [V20]</i>
The agent question	<i>An agent adds breadth over a large trace, and little on one stack frame [V26]. It reads only what the trace and the dump captured [V27]</i>

You will be able to say which half of a session a model can take. [speculative]

[V20] Crash Dump Analysis · [V26] [V27] AI Agents as Debugging Operators

Topic seven is failure analysis, in two halves. The first is the crash dump. Every tool in topics one to six needs the process to be alive when you look. Post-mortem work begins where that stops being true. Tracing shows you a running system. A dump is all you get once it is dead. It is a photograph, not a recording, and you reason backward from it. The second half is what an agent changes. My honest position is that it is tool-specific. Reading a forty thousand line trace is breadth-shaped work, and a model is good at it. Reasoning about one optimised-away local is depth-shaped, and a model is unremarkable. [V26] Knowing which is which is the skill. An agent reads the evidence a tool already produced. It acquires none of its own. [V27] [V20] [V21]

20 20. Topic 8 — Leaks in the abstraction

On screen:

Part II · Topic 8

The stance

Your program steps outside the conditions the abstraction covers. "The leak is not the abstraction failing; it is the program leaving the scope." [V4]

What you take away

Your own list of scope clauses a program stepped outside of [speculative]

The pace

No fixed week per topic. Three weeks are reviews built from your submissions [speculative]

By then your list has entries like a write that returned before the data was safe. [speculative]

[V4] Where the Abstraction Leaks — author's notes.

Topic eight is the synthesis lecture at the end of term. It returns to the idea we started this part with. A leak is not a broken abstraction. It is your program stepping outside the conditions it was defined for, and the leak is information about the layer below. By then you will have a personal list of scope clauses you watched a program walk out of. A write that returned before the data was safe. A memory figure measuring something other than memory. One last thing, about pacing. There is no fixed week per topic. Three weeks are review lectures built from what your submissions got wrong, so I cannot know in September what they contain. You get the topics and the order. The pace follows the cohort. [V4] [speculative]

21 21. How you are assessed

On screen:

Part III

The rules, the dates, and the argument behind them.

Part three. Six minutes, and this is the part you will want to write down. I am going to give you the rules, then the exact dates, then the reasoning. If the reasoning sounds like it belongs in an orientation rather than a syllabus, that is the point I made in part one. How a course measures you tells you what it thinks the subject is.

22 22. Two components: course credit and exam

On screen:

Part III · Components

course credit (zápočet)	<i>You earn it by submitting at least one of the three homework assignments. You do not have to pass one. The checker can fail your submission, and it still earns the course credit. You must pass the exam as well</i>
exam (zkouška)	<i>No AI of any kind in the room — no model you bring, no agent harness. Passing an assignment shrinks it [speculative]</i>

Both rules above are specific to PB158. [speculative] I give the shrinking mechanism, and the reason for the AI rule, later in this part.

You earned a course credit before and you sat an exam before, so I am not going to explain the two words. IS MUNI calls them zápočet and zkouška. Look for those words when you open your own record. I am going to give you the two things that are specific to PB158. First, the course credit. You earn it by submitting at least one of the three homework assignments. Submitting, not passing. The checker can fail your submission, and it still earns the course credit. You must pass the exam as well. Second, the exam. You write it in a classroom, and there is no AI in that room. No models you brought, no harnesses. I know how that sounds. Hold the thought. I come back to it later in this part. I also give you the mechanism that shrinks the exam when you pass an assignment. [speculative]

23 23. Three assignments, and the real dates

On screen:

Part III · Dates

ASSIGNMENT	SET	DUE, MIDNIGHT	REVIEWED
1	Tue 22 Sep	Fri 2 Oct	Tue 6 Oct
2	Tue 6 Oct	Fri 16 Oct	Tue 20 Oct
3	Tue 20 Oct	Fri 30 Oct	Tue 3 Nov

Teams of two to three. Submitted to FI MUNI GitLab at gitlab.fi.muni.cz. Pass or fail, and no points. [speculative]

Nothing is set today. The first assignment arrives next Tuesday, and you have ten days for it. [speculative]

Three assignments this semester. You work in teams of two to three. You submit to the faculty GitLab, at gitlab dot fi dot muni dot cz. Each one is pass or fail, with no points. Read the dates rather than compute them. Assignment one is set next Tuesday, the twenty second of September, and it is due at midnight on Friday the second of October. We review it on Tuesday the sixth. Then the same shape twice more. Set to due is ten days, not two weeks. Plan against two weeks and you lose a weekend. The deadline sits three days before the review, because I need those days to read what you sent. Nothing is set today. [speculative]

24 24. The homework is a loop, not a test

On screen:

Part III · Shape

Resubmit as often as you like until the deadline. A fail costs time and nothing else. [speculative]

What the checker returns

Defects, not remedies. It names what is wrong, never the fix [speculative]

Why not remedies

A critic is normally told to "suggest concrete revisions" [V15]. We drop that half on purpose [speculative]

Push your failures. A passing commit is by construction the one with nothing left that the checker can catch. [speculative]

[V15] Generative Actor Critic — author's notes.

Now the shape of the homework, because it changes how you should work. You may re-submit as often as you like until the deadline. A fail before the deadline costs time and nothing else. So this is not a test with retries. It is a loop. You submit, something tells you what is wrong, you fix it, you submit again. One deliberate choice inside it. The checker returns defects, never remedies. It will say your program loses its output when the output is redirected. It will not say change line forty two. In the engineering version of this pattern the critic does both. [V15] We drop the second half, because a checker that hands you the fix has done the assignment for you. [speculative] And one con-

sequence. A team that iterates to green has, by the deadline, removed every defect the checker tests for. [speculative] So the teaching material is the sequence of attempts, not the pass. [speculative]

25 25. What passing an assignment buys you

On screen:

Part III · Reward

1	<i>The exam has two parts</i>	<i>An irreducible core every student sits unaided, and a removable part [speculative]</i>
2	<i>A pass removes its subject</i>	<i>Passing an assignment removes that subject from your exam. Fixed, not chosen by you [speculative]</i>
3	<i>The whole team gets it</i>	<i>The submission is the team's, and the reduction applies to every member [speculative]</i>

A checker you may retry without limit is a target by construction, so it stops measuring you. "When a measure becomes a target, it ceases to be a good measure" [V16]. That is why the unaided core stays. [inference]

[V16] Goodhart's Law — author's notes.

This is the part that pays you. The exam has two parts. An irreducible core that every one of you sits, and a removable part. Passing an assignment removes that assignment's subject matter from your exam. It is fixed, not chosen by you, and you know on the day it is set exactly what passing buys. So it is a smaller syllabus, not merely a shorter paper. Pass all three and only the core is left. The submission is the team's, so the reduction applies to every member. Yes, your exam depends on your teammates. Hear that on day one rather than in January. Why keep a core? Goodhart's law. When a measure becomes a target it stops measuring, and an unlimited retry checker is a target. The core is unaided, and everybody sits it. [V16] [speculative]

26 26. An AI grades you, and a human is on the loop

On screen:

Part III · The grader

An AI grades both the exam and the homework. A human stays on the loop. [speculative]

What that means

Nobody checks a grade before you get it. A human catches a wrong result afterwards: "the action does not wait for the human. It proceeds by default" [V10]

The asymmetry

You may not use AI in the exam, and an AI grades you. Named here rather than found later [speculative]

[V10] Human on the Loop — author's notes.

An AI grades both the exam and the homework. A human is always on the loop. Let me say precisely what on the loop means, because it is weaker than it sounds. The defining feature is negative. The action does not wait for the human. It proceeds by default, and the human's role is to catch it, not to authorise it. Nobody checks before a fail is a fail. Now the thing you were all going to work out eventually. You may not use AI in the exam, and an AI grades you. I would rather say that myself than have you find it in week six. They are different tasks. The exam measures what you can do unaided. [V10] [speculative]

27 27. How this arrangement fails

On screen:

Part III · Failure

<i>Vigilance decays</i>	<i>Mackworth's watchers hunted rare double jumps in a pointer stepping once a second, for two hours [V24]. Detection fell from about 85% to about 75% in 30 minutes [V11] [inference]</i>
<i>Review becomes ritual</i>	<i>Gate everything and the reviewer clicks through: "a rubber stamp with an audit trail" [V12] [inference]</i>
<i>A critic invents faults</i>	<i>Where models started at $\geq 75\%$ task accuracy, "The critic, primed to find errors, invented them." [V14] [inference]</i>

So tell us when a result is wrong. "the human's job is to adjudicate a raised alarm" [V13]

[V11] [V12] [V13] [V24] Human on the Loop · [V14] Generative Actor Critic

This is the first system the course asks you to look at rather than trust, and it is the one grading you. So here is how it fails. Bainbridge named the general problem in nineteen eighty three. The better the automation, the worse the monitor's job becomes. Mackworth measured it. His observers watched a pointer step once a second for two hours, and had to catch its rare double jumps. Detection fell from about eighty five percent to about seventy five within thirty minutes. Second. Make a human sign off everything and you train them to click through. That is a rubber stamp with an audit trail. Third, and this one bites good work. On the tasks where the models started at seventy five percent accuracy or better, a critic primed to find errors invented them. The repair is structural. Tell us when a result is wrong. A human adjudicates a raised alarm far better than they spot one. [S8] [V11] [V24] [V12] [V14] [V13]

28 28. What this course is betting

On screen:

Part III · Honesty

The bet: unlimited retries, with two counterweights — a checker that names defects, never remedies, and an unaided exam core. [speculative]

Three published results argue against it. [inference]

Retries [S6]	<i>A team pushes one-line edits to see which test flips [speculative]. Limits and regression penalties cut that by up to 85%</i>
Feedback [S18]	<i>Rapid feedback leads some students to aim at the tests, not the program</i>
Guidance [S22]	<i>Under minimal guidance a learner with low prior knowledge learns less than under strong guidance</i>

Edwards' counterweight was student-written tests [S17]. Ours are the two above. [speculative]

[S6] <https://dl.acm.org/doi/10.1145/3591210> [S17] <https://web-cat.org/publications/pos27-Edwards.pdf> [S18] <https://valerio-terragni.github.io/assets/pdf/luxton-ace-2023.pdf> [S22] https://itgs.ict.usc.edu/papers/Constructivism_KirschnerEtAl_EP_06.pdf

Here is the bet, and this is the slide I would most want a colleague to attack. We keep unlimited retries. Two counterweights carry the risk. The first is a checker that names defects and never remedies. The second is an exam core that every one of you sits unaided. [speculative] Now the evidence against it. Lawrence and colleagues measured the case against us. A team pushes a one line edit every few minutes to see which test flips. Submission limits and regression penalties together cut that behaviour by up to eighty five percent. Students also prefer the limits. Luxton-Reilly and colleagues name the behaviour. With rapid feedback, some students aim at passing the test cases rather than improving the program. Kirschner, Sweller and Clark compared minimal guidance with strong guidance. Under minimal guidance a learner with low prior knowledge learns less of the material than under strong guidance. Their finding is about learners with low prior knowledge. Reading this room as that case is my judgement, not theirs. [speculative] Edwards built unlimited submission in two thousand three with a counterweight of his own. He assessed a student's program only by the tests the student had written. We keep the unlimited half and substitute our two. If the bet is wrong, it is wrong here. [S6] [S18] [S22] [S17] [speculative]

29 29. Where to keep learning

On screen:

Part IV · Reading

1	Julia Evans	<i>jvns.ca — debugging and tracing from the outside in, including why strace fails in Docker [S12]</i>
2	Raymond Chen	<i>The Old New Thing — decades of why Windows is the way it is, still publishing [S14]</i>
3	Brendan Gregg	<i>flame graphs, eBPF, and the tracing overhead figures this course quotes [S13]</i>
4	Oxide and Friends	<i>Bryan Cantrill co-created DTrace, one of the two tracing traditions here [S16]</i>

Pick one. The handout carries the full annotated list with URLs.

[S12] <https://jvns.ca/blog/2020/04/29/why-strace-doesnt-work-in-docker/> · [S13] <https://www.brendangregg.com/> · [S14] <https://devblogs.microsoft.com/oldnewthing/> · [S16] <https://oxide.computer/podcasts/oxide-and-friends>

One minute on reading, and I kept this to four. Julia Evans writes about debugging and tracing better than anyone. Her piece on why strace does not work inside Docker is exactly the failure you will hit in the first lab. Raymond Chen has written The Old New Thing since two thousand three. It is the best single source on why Windows is shaped the way it is. Brendan Gregg is where this course's tracing overhead numbers come from. Oxide and Friends is a podcast by Bryan Cantrill, who co-created DTrace. Pick one and actually follow it. The full annotated list is in the handout, so do not photograph this slide. [S12] [S13] [S14] [S16] [speculative]

30 30. What must work on your Windows machine

On screen:

Part V · Windows

TOOL	WHAT FOR
Visual Studio Community	<code>cl.exe</code> , the Windows SDK, PDB symbols. Tick Desktop development with C++
VS Code + C/C++ extension	the editor, on both platforms
Sysinternals Suite	Process Monitor and Process Explorer
NtTrace	native API tracing, the <code>strace</code> counterpart on Windows
WinDbg	crash dumps, for topic 7
WSL2	how you reach the Linux side from Windows
Python and uv	"An extremely fast Python package and project manager, written in Rust" [S21]

Microsoft native, by decision. No MinGW and no MSYS2, because WinDbg wants MSVC symbols. [speculative]

The Windows list. Visual Studio Community, and you must tick Desktop development with C plus plus or you will not get the compiler. Community is free for classroom use. Build Tools is the wrong package, because it needs a Visual Studio licence. VS Code with the C and C plus plus extension. The Sysinternals Suite for Process Monitor and Process Explorer. NtTrace, the `strace` counterpart. WinDbg for topic seven. WSL2, because that is how you reach the Linux side. And Python with uv, a very fast package manager written in Rust. Use it instead of pip. One decision worth stating. This is the Microsoft native toolchain by choice, because WinDbg wants MSVC symbols. [S21] [speculative]

31 31. What must work on your Linux machine

On screen:

Part V · Linux

TOOL	WHERE
VS Code	<code>code.visualstudio.com</code> , same editor as the Windows side
gcc, make, libc headers	your distribution's build-essential equivalent
strace and gdb	your distribution's own packages
Python and uv	<code>docs.astral.sh/uv</code> [S21]

The course **dev container** carries all of this. A dev container is "a running container with a well-defined tool/runtime stack and its prerequisites" [S10]

Faculty machines exist for both platforms, so you are not blocked if your laptop refuses. [speculative]

[S10] Development Containers — containers.dev. <https://containers.dev/> · [S21] uv — Astral. <https://docs.astral.sh/uv/>

The Linux list is shorter. VS Code, the same editor. Your distribution's build essential equivalent for gcc, make and the C library headers. strace and gdb from your own package manager. Python and uv again. The easy route is the course dev container, which carries all of it preinstalled. A dev container is a running container with a defined tool stack and its prerequisites. A file in the repository describes it, so everyone gets the same environment. If you are on Windows, the dev container is how you reach Linux, through WSL2. And if your own laptop refuses any of this, the faculty provides machines for both platforms. Do not spend a week fighting your own hardware. Tell us instead. [S10] [S21] [speculative]

32 32. Two things that cost an evening

On screen:

Part V · Warnings

1

Visual Studio Community is large

Microsoft's figure is 20 to 50 GB for a typical installation [S19]

2

`cl.exe` needs its own shell

"Even a simple Visual Studio installation can set twenty or more environment variables" [S20]

Use the Developer Command Prompt. An ordinary shell says "not recognised", which reads as a broken install rather than a missing environment. [inference]

[S19] Visual Studio 2026 system requirements · [S20] MSVC from the command line — Microsoft Learn. <https://learn.microsoft.com/en-us/visualstudio/releases/2026/vs-system-requirements>

Two warnings, because both cost people an evening every year. First, Visual Studio Community is not a small download. Microsoft's own figure is twenty to fifty gigabytes for a typical installation, and the full range goes up to two hundred and ten. Check your free space first. Install it before the first lab rather than tonight, because the first assignment does not need it. Second, `cl dot exe` does not work from an ordinary PowerShell. Microsoft says a simple installation can set twenty or more environment variables, which is why they ship a Developer Command Prompt. In an ordinary shell the error just says not recognised, and every year somebody reinstalls Visual Studio Community because of it. [S19] [S20] [inference]

33 33. Summary — what follows from this

On screen:

Lecture 1 · Close

1

Nothing is graded on recall

The exam core measures what you can do unaided. Homework is a loop [speculative]

2

Submit the attempt you doubt

The review lectures are built from your failures [speculative]

3

Check us

The grader has named failure modes [V11] [V12]. You raise the alarm. A human adjudicates it [V13]

Now the second lecture: three lines of Python whose output disappears.

[V11] [V12] [V13] *Human on the Loop*

Three things that follow. First, nothing here is graded on recall. The exam core measures what you can do unaided, and the homework is a loop you iterate against. Revision in the ordinary sense will not help you much. Second, submit the attempt you are unsure about. Your failed attempts are what the review lectures are made of, and a repository of clean passing commits teaches nobody anything. Third, check us. The grading arrangement has named failure modes. I have shown you three. Raising one is your job as much as mine. That is the orientation. Short break, then the actual subject, with three lines of Python whose output vanishes. [speculative] [inference]

34 34. References

On screen:

- [S1] *The misunderstood limits of folk science: an illusion of explanatory depth* — Rozenblit & Keil, *Cognitive Science* 26 (2002). <https://cogdevlab.yale.edu/sites/default/files/files/rozenblit%20&%20keil%20%202002.pdf>
- [S2] *Classroom demonstrations: learning tools or entertainment?* — Crouch, Fagen, Callan & Mazur, *Am. J. Phys.* 72 (2004). <https://works.swarthmore.edu/fac-physics/203/>
- [S3] *Teaching Operating Systems with FreeBSD through Tracing, Analysis, and Experimentation* — Neville-Neil et al., *USENIX ;login:* (2017). https://www.usenix.org/system/files/login/articles/login_spring17_03_neville-neil.pdf
- [S4] *On teachers working with large-language-model grading in classrooms* — Tian et al. (2025), *arXiv:2506.07955*. Recorded title contradicted on re-reading, and no exact title confirmed in this build. <https://arxiv.org/abs/2506.07955>
- [S5] *Lessons Learned from 30 Years of MINIX* — Tanenbaum, *CACM* (2016). <https://www.cs.vu.nl/~ast/Publications/Papers/cacm-2016.pdf>
- [S6] *On submission limits and regression penalties* — Lawrence, Foss & Urazova, *ACM TOCE* (2023). <https://dl.acm.org/doi/10.1145/3591210>
- [S7] *The Keller Plan and the Personalized System of Instruction* — Eyre, *Behav. Analyst Today* 8(3) (2007). <https://files.eric.ed.gov/fulltext/EJ800986.pdf>
- [S8] *Ironies of Automation* — Bainbridge, *Automatica* 19(6) (1983). <https://gwern.net/doc/sociology/technology/1983-bainbridge.pdf>

- **[S9]** MU Study and Examination Regulations — Masaryk University, official notice board, current edition. <https://www.muni.cz/en/about-us/official-notice-board/mu-study-and-examination-regulations>
- **[S10]** Development Containers — containers.dev. <https://containers.dev/>
- **[S11]** Create a Dev Container — Visual Studio Code documentation. <https://code.visualstudio.com/docs/devcontainers/create-dev-container>
- **[S12]** Why strace doesn't work in Docker — Julia Evans, jvns.ca (2020). <https://jvns.ca/blog/2020/04/29/why-strace-doesnt-work-in-docker/>
- **[S13]** brendangregg.com — Brendan Gregg. <https://www.brendangregg.com/>
- **[S14]** The Old New Thing — Raymond Chen, Microsoft Developer Blogs. <https://devblogs.microsoft.com/oldnewthing/>
- **[S15]** LWN.net — weekly Linux kernel reporting. <https://lwn.net/>
- **[S16]** Oxide and Friends — Bryan Cantrill and Adam Leventhal. <https://oxide.computer/podcasts/oxide-and-friends>
- **[S17]** Web-CAT project paper, pos27-Edwards.pdf — Stephen H. Edwards, Virginia Tech. <https://web-cat.org/publications/pos27-Edwards.pdf>
- **[S18]** On rapid feedback and student submission behaviour — Luxton-Reilly, Tempero & Arachchilage, ACE '23. <https://valerio-terragni.github.io/assets/pdf/luxton-ace-2023.pdf>
- **[S19]** Visual Studio 2026 system requirements — Microsoft Learn. <https://learn.microsoft.com/en-us/visualstudio/releases/2026/vs-system-requirements>
- **[S20]** Using the MSVC toolset from the command line — Microsoft Learn. <https://learn.microsoft.com/en-us/cpp/build/building-on-the-command-line>
- **[S21]** uv — a Python package and project manager — Astral. <https://docs.astral.sh/uv/>
- **[S22]** Why Minimal Guidance During Instruction Does Not Work — Kirschner, Sweller & Clark, Educational Psychologist 41(2) (2006). https://itgs.ict.usc.edu/papers/Constructivism_KirschnerEtAl_EP_06.pdf
- **[V1] [V2] [V3]** Observability vs Monitoring — author's working notes.
- **[V4] [V5] [V6]** Where the Abstraction Leaks — author's working notes.
- **[V7]** Sockets and Network I/O — author's working notes.
- **[V8]** Windows and Linux place the stable syscall boundary at different layers — author's working notes.
- **[V9] [V23]** Observing a Running System — author's working notes.
- **[V10] [V11] [V12] [V13] [V24]** Human on the Loop — author's working notes.

- **[V14] [V15]** *Generative Actor Critic as an Agentic Orchestration Pattern* — author's working notes.
- **[V16]** *Goodhart's Law* — author's working notes.
- **[V17]** *Virtual Memory and the Page Cache* — author's working notes.
- **[V18]** *Memory Protection* — author's working notes.
- **[V19]** *Processes, Threads and Scheduling* — author's working notes.
- **[V20]** *Crash Dump Analysis on Windows and Linux* — author's working notes.
- **[V21] [V26] [V27]** *AI Agents as Debugging Operators* — author's working notes.
- **[V22] [V25]** *Every Tracing Mechanism Trades Observation Cost for a Blind Spot* — author's working notes.

The handout PDF carries the full reference list, and a one-page summary of every [V] source with its own external references.

The full source list. Every claim in this lecture points at one of these. The handout uses the same numbering, and its appendix summarises each of the working notes. If you want to check something I said, that is where to start.